Unleashing the Power of CLIP as a General Teacher for Multi-Task Learning

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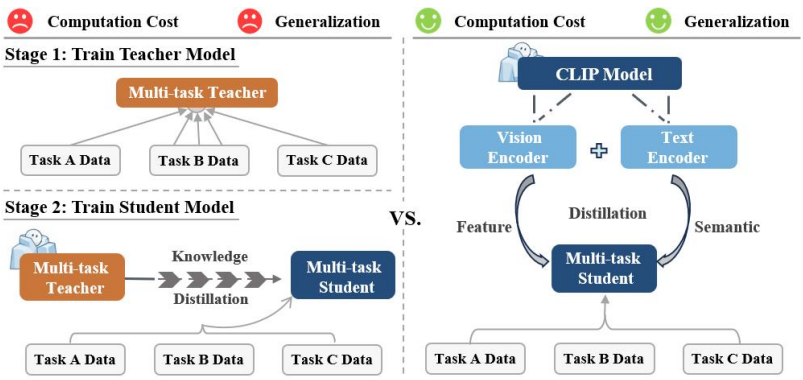
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I. Motivations & Contributions

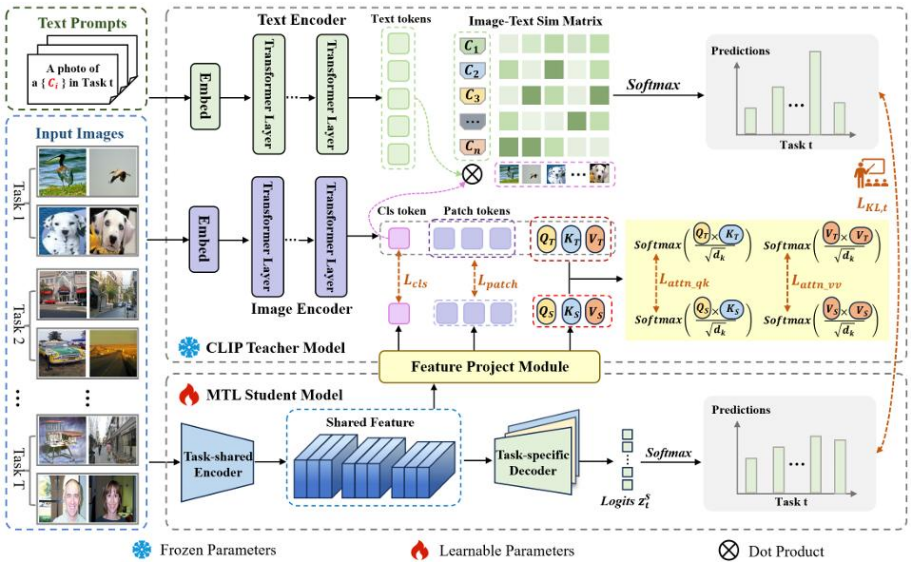
This paper addresses the limitations of existing MTL-KD, where task-specific teachers require costly downstream training and exhibit limited generalization.

Contributions:

- We introduce CLIP as a general and task-agnostic teacher for MTL.
- We propose CMKD, a dual-path framework for transferring shared visual representations and task-specific semantic knowledge.
- State-of-the-Art Performance



II. Method



CMKD uses frozen CLIP as a general teacher and transfers knowledge to the MTL student through two paths:

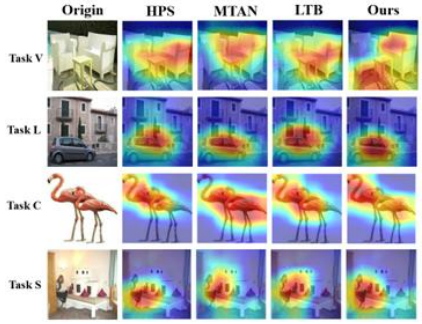
- Representation Distillation: GF, LF, and AF projectors align **global, local, and attention features** between CLIP and the shared student encoder.
- Semantic Distillation: CLIP's **image-text logits** supervise task-specific student heads.
- Objective: Task loss + feature distillation + semantic distillation.

III. Experiment

☐ Performance Comparison & Visualization Analysis

Table 1: Comparisons on VLCS

	Domain-V	Domain-L	Domain-C	Domain-S	Avg ↑
Single-Task	84.32	75.40	100.0	78.70	84.60
Student(MoLA)	83.05	77.01	99.00	80.00	84.76
MGDA	82.63	78.61	100.0	75.65	84.22
PCGrad	85.59	75.40	100.0	80.87	85.47
CAGrad	86.86	75.94	100.0	83.48	86.57
HPS	84.75	74.33	99.00	82.61	85.17
Cross-Stitch	**88.98**	73.26	100.0	78.26	85.13
MMoE	85.59	73.26	100.0	77.39	84.06
CGC	86.44	78.07	98.00	76.52	84.76
LTB	87.71	72.19	100.0	81.30	85.30
MTAN	86.44	75.40	99.00	82.17	85.75
Teacher	86.44	68.45	100.0	81.30	84.05
CMKD(Ours)	84.75	**80.21**	100.0	**83.91**	**87.22**



☐ Shared Representation Generalization

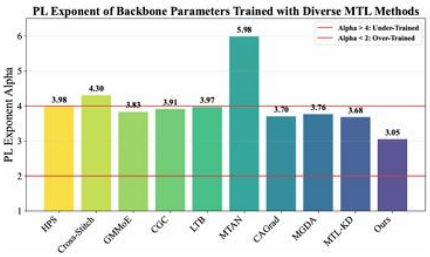


Figure 7: Comparison of PL exponent alpha for backbone parameters trained with diverse MTL methods on VLCS dataset.